Deaths from Lightning, Heat Stroke and Floods: A State Typology for Early-Warning Priorities

Machine Learning | Disaster Management | Climate Analysis | India

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Abstract

This case-study analyses State/Union Territory (UT)-level mortality profiles associated with lightning, heat stroke and floods in India, and examines the statistical association between summer maximum temperature and heat-stroke mortality. The mortality analysis uses the National Crime Records Bureau (NCRB) Accidental Deaths and Suicides in India (ADSI) State/UT data for 2001–2014, combined with 2011 Census population denominators to calculate approximate deaths per lakh population. State-level hazard-rate profiles are standardised and analysed using K-Means clustering, PCA visualisation, Ward hierarchical clustering and Isolation Forest outlier detection. The supplied clustering evaluation records its highest tested silhouette score at k=4 (0.4971), with SSE 27.4265. The four clusters contain 19, 5, 7 and 4 State/UT records respectively. A separate linear regression uses summer mean maximum temperature as a single predictor of heat-stroke death rate, with 2001–2011 observations for training and 2012–2014 for testing. On the held-out test set (78 State-year observations), the model obtains MAE 0.0850, RMSE 0.1352 and R^2 0.0363, modestly outperforming the training-mean baseline on these metrics. These results support exploratory State-level risk profiling, but do not establish causation or demonstrate a strong predictive relationship. Findings are subject to the fixed-population denominator, historical data coverage, geographic harmonisation and temperature-grid aggregation limitations.

Keywords: Disaster mortality; lightning; heat stroke; floods; India; K-Means; PCA; hierarchical clustering; Isolation Forest; linear regression; summer maximum temperature; early-warning systems.

Article type: Case Study. The work examines a defined real-world setting—Indian States/UTs—using historical mortality and climate data to produce evidence for targeted disaster preparedness and early-warning priorities.

1. Introduction

Lightning, heat stroke and floods are distinct hazards that can produce substantial human losses, but their relative importance may vary across regions. A uniform national intervention may therefore fail to reflect local hazard profiles. State/UT-level mortality rates offer one way to compare recorded impacts while accounting approximately for differences in population size.

The project addresses three questions: (i) how do recorded lightning, heat-stroke and flood death rates vary across Indian States/UTs? (ii) can States/UTs be grouped into interpretable mortality-profile clusters? and (iii) how strongly is summer mean maximum temperature associated with the heat-stroke death rate in the available matched State-year data?

Objectives. Calculate cause-specific rates per lakh population; identify multivariate mortality profiles; inspect cluster structure and potential outliers; examine the temperature–heat-stroke relationship; and discuss how descriptive evidence may inform targeted heat-action, lightning-warning and flood-preparedness planning.

The work is aligned with SDG 13 (Climate Action), SDG 11 (Sustainable Cities and Communities) and SDG 3 (Good Health and Well-being). These alignments describe the intended application; the analysis does not measure SDG target attainment.

2. Related Work

Disaster-risk management increasingly combines historical impact records, climate observations and spatial analysis to identify exposed populations and support preparedness. In this project, the contribution is a comparative State/UT-level view of three recorded natural-cause mortality categories rather than a detailed event-level hazard model.

K-Means is a partitioning method that groups observations by distance to cluster centroids; feature scaling is important when variables have different numerical ranges. PCA is used here to display multivariate profiles in a lower-dimensional plane, not to prove that the displayed axes represent all variation. Ward's hierarchical method provides a second view of similarity through a dendrogram. The silhouette coefficient evaluates within-cluster cohesion relative to separation from neighbouring clusters. Isolation Forest flags unusual multivariate profiles; such flags are candidates for investigation, not evidence of erroneous data.

For the temperature analysis, ordinary least-squares linear regression estimates a one-predictor association. A held-out temporal test period and a simple training-mean baseline provide a basic check of predictive performance. The short test period and repeated State observations mean that the reported scores should be treated as exploratory.

3. Dataset and Preprocessing

Mortality source. The project uses the NCRB ADSI State/UT-wise distribution of accidental deaths by natural causes, accessed through the Open Government Data (OGD) Platform India. The supplied cleaned mortality table covers 2001–2014 and includes counts for flood, heat stroke, lightning and other natural causes.

Population denominator. The pipeline uses 2011 Census population values for all years to calculate cause-specific rates per lakh: deaths divided by population, multiplied by 100,000. This is explicitly a preliminary approximation, not an annual population series. Rates should therefore not be interpreted as precise year-specific risks.

Climate source. Summer maximum-temperature summaries are derived from IMD daily maximum-temperature GRD files for 2001–2025. The supplied State temperature table contains 725 State-year rows. The regression can only use years overlapping the mortality series, namely 2001–2014.

Spatial and temporal processing. The temperature workflow assigns grid-cell information to State boundaries and produces State-level summer summaries. Results depend on the GRD format assumptions, grid orientation, missing-value handling, boundary matching and the spatial weighting approach. The supplied outputs do not by themselves establish that these assumptions have been independently validated.

Regression sample. The merged regression dataset contains 364 observations across 26 States/UTs for 2001–2014. The supplied test predictions contain 78 observations (26 States/UTs in each of 2012, 2013 and 2014).

4. Methodology and Experimental Setup

4.1 Mortality-rate profiles. Cause-specific rates per lakh population are used to represent each State/UT. The clustering output contains 35 State/UT profiles and three hazard-rate features: lightning, heat stroke and flood. The profile table appears to represent State-level summary rates; interpretation should be tied to the actual aggregation implemented in the pipeline.

4.2 Scaling and clustering. Features are standardised before distance-based clustering. K-Means is evaluated for k=2 to k=8 using SSE and silhouette score. The selected k is the tested value with the highest silhouette score; the elbow curve and substantive interpretability should also be considered. PCA provides a two-dimensional visualisation. Ward hierarchical clustering is used as a complementary similarity view.

4.3 Outlier detection. Isolation Forest is applied to the scaled hazard-rate profiles. Its output labels profiles as normal or potential outliers. A potential outlier may reflect a genuinely distinctive hazard profile, small denominators, reporting variation or data-processing issues; it requires contextual review.

4.4 Regression. Linear Regression models heat-stroke deaths per lakh as a function of summer mean maximum temperature. The chronological split uses 2001–2011 for training and 2012–2014 for testing. Performance is reported using MAE, RMSE and R^2, and compared with a baseline that predicts the training mean. The model is univariate and observational.

Software. The project outputs were generated by Python scripts using common data-science libraries including pandas, NumPy, scikit-learn and plotting tools. Exact package versions and hardware details were not present in the supplied outputs and are therefore not reported.

5. Results and Discussion

5.1 Cluster selection. Among the tested values, k=4 has the highest silhouette score (0.4971) and SSE 27.4265. Silhouette scores are 0.4039, 0.4581, 0.4971, 0.4639, 0.4379, 0.4169 and 0.4228 for k=2 through k=8 respectively. This supports k=4 as the best candidate among those tested, not as a universally optimal or definitive partition.

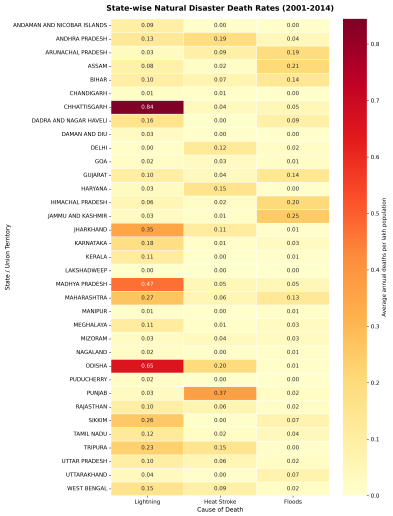


Figure 1. State-wise hazard mortality-rate heatmap (per lakh population).

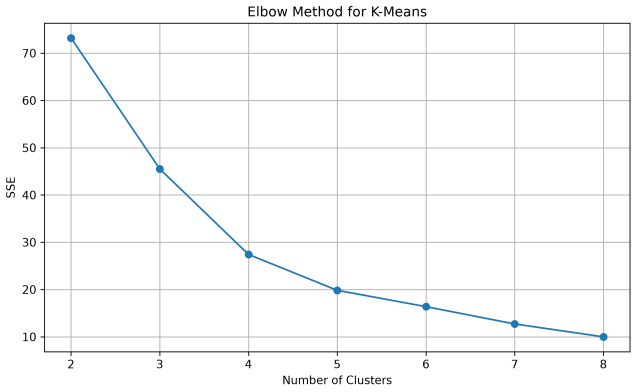


Figure 2. K-Means within-cluster sum of squared errors (SSE) for tested k values.

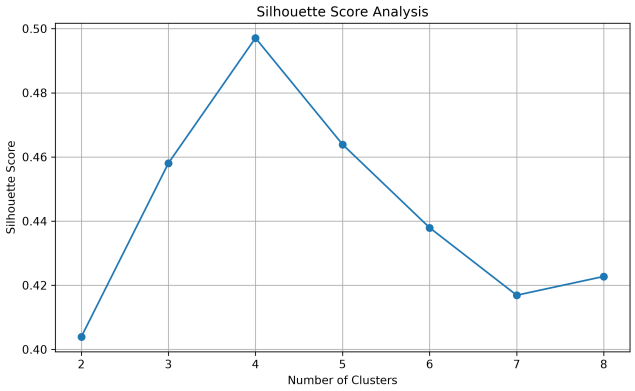


Figure 3. Silhouette scores for k=2 to k=8; the maximum occurs at k=4.

5.2 Cluster profiles. The 35 State/UT profiles are distributed as follows. Cluster 0 contains 19 profiles and has the lowest mean heat-stroke and flood rates in this partition (0.0175 and 0.0242 per lakh). Cluster 1 contains five profiles and has the highest mean heat-stroke rate (0.1966 per lakh). Cluster 2 contains seven profiles and has the highest mean flood rate (0.1795 per lakh). Cluster 3 contains four profiles and has the highest mean lightning rate (0.5811 per lakh). These labels are descriptive summaries of cluster means, not official hazard classifications.

Cluster	States/UTs	Mean lightning	Mean heat stroke	Mean flood
0	19	0.0813	0.0175	0.0242
1	5	0.0844	0.1966	0.0178
2	7	0.0967	0.0426	0.1795
3	4	0.5811	0.0995	0.0304

Cluster membership. Cluster 1 comprises Andhra Pradesh, Delhi, Haryana, Punjab and Tripura. Cluster 2 comprises Arunachal Pradesh, Assam, Bihar, Gujarat, Himachal Pradesh, Jammu and Kashmir and Maharashtra. Cluster 3 comprises Chhattisgarh, Jharkhand, Madhya Pradesh and Odisha. Cluster 0 contains the remaining 19 profiles listed in the supplied clustering table.

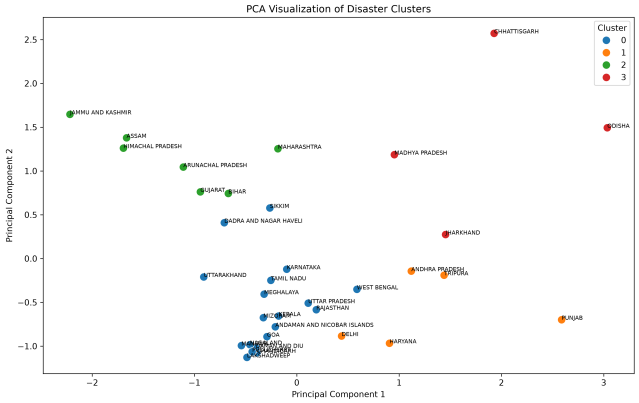


Figure 4. PCA projection of State/UT mortality profiles, coloured by K-Means cluster.

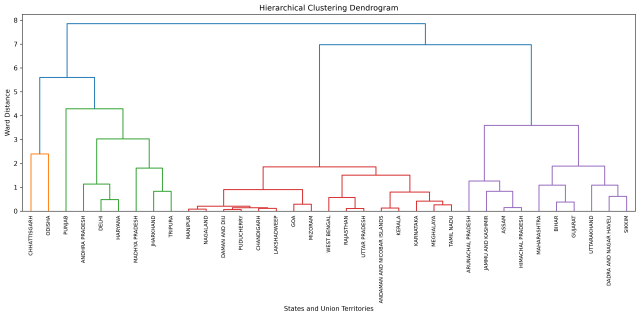


Figure 5. Ward hierarchical clustering dendrogram for the State/UT profiles.

5.3 Potential outliers. Isolation Forest flags 11 profiles: Andhra Pradesh, Arunachal Pradesh, Assam, Chhattisgarh, Himachal Pradesh, Jammu and Kashmir, Jharkhand, Madhya Pradesh, Maharashtra, Odisha and Punjab. These should be treated as unusual profiles under the fitted detector, not automatically as errors or high-risk jurisdictions.

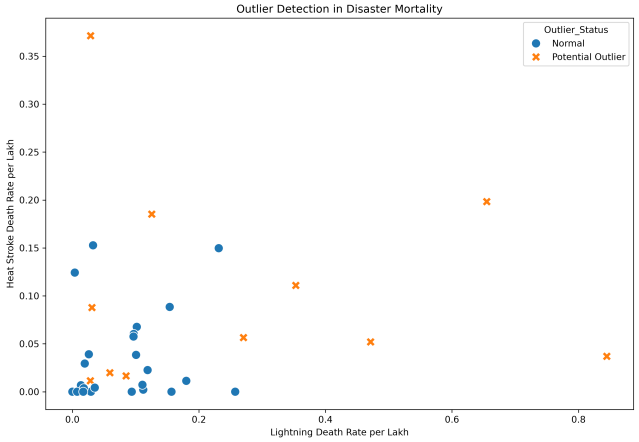


Figure 6. Isolation Forest outlier flags against lightning and heat-stroke mortality rates.

5.4 Temperature and heat-stroke regression. The linear regression achieves MAE 0.0850, RMSE 0.1352 and R^2 0.0363 on the 2012–2014 test period. The training-mean baseline obtains MAE 0.0914, RMSE 0.1405 and R^2 –0.0420. Thus, the linear model performs slightly better than the baseline on the reported test metrics, but its R^2 indicates that it explains only a small proportion of test-set variation. The fitted equation estimated from the supplied 2001–2011 training data is: heat-stroke rate = -0.1570 + 0.00668 × summer mean maximum temperature (°C).

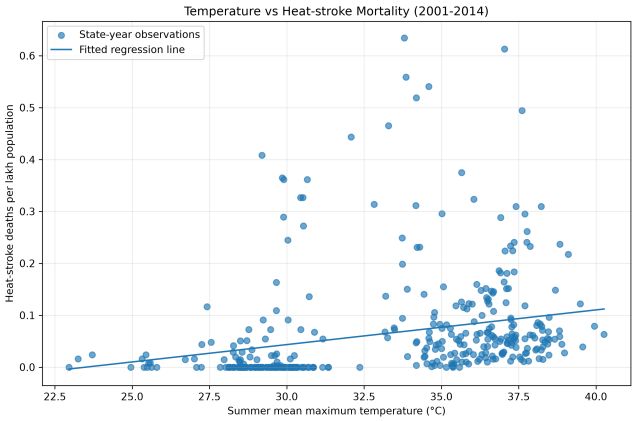


Figure 7. Summer mean maximum temperature versus heat-stroke mortality rate.

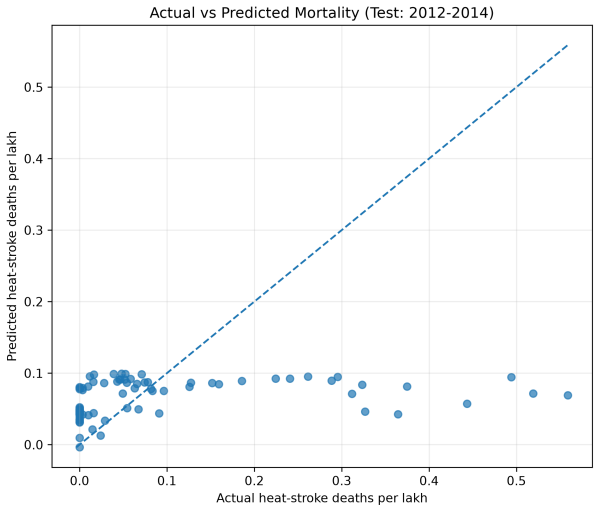


Figure 8. Actual and predicted heat-stroke mortality rates in the 2012–2014 test set.

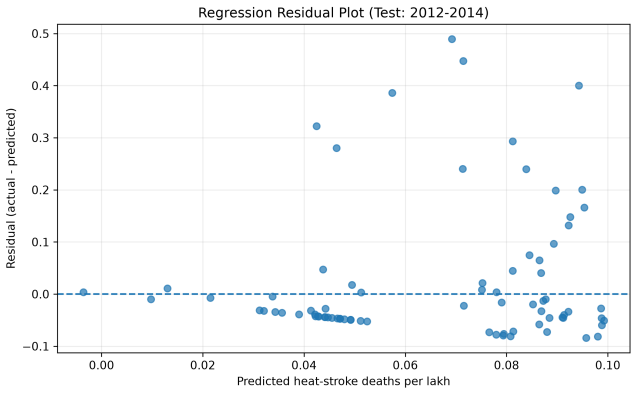


Figure 9. Test-set residuals (actual minus predicted) against predicted rates.

The positive fitted slope is consistent with a positive association in the training sample, but the low test R^2 means temperature alone is not a strong explanation of State-year heat-stroke mortality in this experiment. Humidity, heatwave duration, occupational exposure, age structure, urbanisation, healthcare access, adaptation and reporting practices are not included in the one-predictor model. Repeated observations for the same State/UT also mean that test rows are not fully independent.

6. Conclusion and Future Work

This exploratory case study demonstrates a reproducible analytical route for comparing State/UT mortality profiles across lightning, heat stroke and floods. In the supplied clustering results, four groups provide the highest tested silhouette score; their mean hazard profiles differ, with one cluster showing relatively higher lightning rates, another higher flood rates and another higher heat-stroke rates. Outlier detection identifies a set of profiles that merit contextual inspection.

The regression results show only modest improvement over a training-mean baseline and a low test R^2. Accordingly, the current analysis should be presented as an examination of association, not as a reliable stand-alone heat-stroke forecasting system and not as evidence of causation. The results can inform further investigation and prioritisation discussions, but operational early-warning decisions require additional validation and hazard-specific evidence.

Future work. Replace the fixed 2011 population denominator with annual/intercensal population estimates; document and validate State/UT boundary harmonisation across historical administrative changes; verify IMD GRD metadata, byte order, grid orientation and missing-value conventions; evaluate area- or population-weighted temperature summaries; add humidity, heatwave duration and socioeconomic/healthcare covariates; compare regularised and nonlinear regression models; use State-aware or panel-data validation; and test clustering stability across years and alternative feature sets.

7. Declarations and References

Data availability. The mortality data are sourced from the NCRB ADSI catalogue on the OGD Platform India. The temperature analysis uses IMD daily maximum-temperature GRD data. Population denominators are from the 2011 Census. The analysis outputs accompany the project repository.

Conflict of interest. The author declares no known competing financial or personal interests relevant to this academic project.

Funding. No external funding was reported for this project.

Ethics. The study uses aggregate, publicly available State/UT-level data and does not analyse individual-level personal records.

AI-tool declaration. If generative AI tools were used for language editing, coding assistance or structuring, the author should disclose the tool and the specific assistance in accordance with the course or venue policy. The author remains responsible for verifying all code, data, interpretations and references.

References

[1] National Crime Records Bureau (NCRB), *Accidental Deaths and Suicides in India (ADSI): State/UT-wise distribution of accidental deaths by natural causes*, Open Government Data Platform India. https://www.data.gov.in/catalog/stateut-wise-distribution-accidental-deaths-natural-causes

[2] India Meteorological Department (IMD), daily gridded maximum-temperature data (1° grid), 2001–2025, as used in the project. Dataset metadata and processing assumptions should be documented with the downloaded files.

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8. Appendix: Additional Results

Appendix A. Cluster evaluation values. The full tested metrics are reproduced below.

k	SSE	Silhouette
2	73.2079	0.4039
3	45.5242	0.4581
4	27.4265	0.4971

k	SSE	Silhouette
5	19.8429	0.4639
6	16.3871	0.4379
7	12.7240	0.4169
8	9.9989	0.4228

Appendix B. Selected highest observed State/UT rates in the clustering profile table.

Lightning: CHHATTISGARH: 0.845; ODISHA: 0.655; MADHYA PRADESH: 0.471; JHARKHAND: 0.353; MAHARASHTRA: 0.270 deaths per lakh.

Heat stroke: PUNJAB: 0.371; ODISHA: 0.198; ANDHRA PRADESH: 0.185; HARYANA: 0.153; TRIPURA: 0.150 deaths per lakh.

Flood: JAMMU AND KASHMIR: 0.250; ASSAM: 0.205; HIMACHAL PRADESH: 0.203; ARUNACHAL PRADESH: 0.191; GUJARAT: 0.141 deaths per lakh.

Appendix C. Interpretation and data cautions. (i) The clustering table contains 35 profiles, while the merged regression dataset contains 26 States/UTs; the exact geographic coverage differs between analysis stages and should be explained in a final reproducibility note. (ii) State/UT names and boundaries change over time; historical harmonisation may affect comparisons. (iii) A fixed 2011 population denominator is not an annual denominator. (iv) Grid-cell assignment and aggregation can affect State temperature estimates. (v) The regression test period is only three years. (vi) Cluster IDs are arbitrary labels and should not be interpreted as rankings.

Prepared from the output files supplied with the project ZIP. Numerical findings in this report are drawn from those files; no missing values or unreported experimental details have been fabricated.